

Exam in Numerical Analysis FMN050–OLD, 050818

Gustaf Söderlind

IMPORTANT! — EXAM FOR OLD STUDENTS ONLY

THIS EXAM IS ONLY FOR STUDENTS WHO TOOK THE FMN050 COURSE IN 2004 OR EARLIER. WRITE CLEARLY ON YOUR EXAM FOLDER THAT YOU HAVE TAKEN THE “FMN050 OLD EXAM.”

The exam lasts 14:00–17:00. Preliminary requirements: for **grade 3**, 15–19 points are required; **grade 4**: 20–24 p, **grade 5**: 25–30 p.

You are allowed to use a pocket calculator, the course compendium (Führer & Schroll) and printouts of slides, but no other material of any kind. Please answer the problems in Swedish or English.

Problems

A. Condition and Error analysis

1. Let $F(x) = Ax$ for a nonsingular matrix A and $x = (x_1, x_2, x_3)^T \neq 0$. Suppose that $y = (y_1, y_2, y_3)^T \neq 0$ has an error of δy , and consider the problem of solving $F(x) = y$.

a) Using the information above, give a bound for the absolute error δx in the solution x to the problem $F(x) = y$. (Explain your notation.) **(2p)**

b) Give a bound for the relative error in the solution x to the problem the problem above. (Explain your notation.) **(2p)**

c) Define the condition number for solving the problem above. (Explain your notation.) **(1p)**

Solution. In a) we have $Ax = y$ with solution $x = A^{-1}y$. For the perturbations it then holds $\delta x = A^{-1}\delta y$. The bound is $\|\delta x\| \leq \|A^{-1}\| \cdot \|\delta y\|$.

In b) we have the standard relative error bound

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa[A] \frac{\|\delta y\|}{\|y\|},$$

where $\kappa[A] = \|A^{-1}\| \cdot \|A\|$ is the relative condition number.

B. Interpolation and Least squares

1. In theory, the braking distance s for a car travelling at speed v is $s(v) = cv^2$. If the car decelerates at a constant rate a m/s², then according to classical mechanics, we have

$$-2as = v^2.$$

(Note that a is negative.) The car's brakes were tested by braking to full stop as fast as possible, five times from a speed of 90 km/h, equivalent to 25 m/s. The braking distance, measured in meters, is found in Table 1 below. Determine the deceleration a using the least squares method. **(2p)**

v	25	25	25	25	25
s	32	31	34	32	33

Table 1: Data for problem **B1**.

Solution. The overdetermined system $-2as = v^2$ becomes

$$\begin{bmatrix} -64 \\ -62 \\ -68 \\ -64 \\ -66 \end{bmatrix} a = \begin{bmatrix} 625 \\ 625 \\ 625 \\ 625 \\ 625 \end{bmatrix}.$$

Solving using the normal equations yields $a = -9.6355$. (This is equivalent to computing the average of the five different measurements of a .)

C. Integration

1. Compute the integral

$$\int_0^1 xe^{-x} dx$$

using Romberg's method to an accuracy of 10^{-5} . For full credits you need to show by some argument that the requested accuracy has been attained. Give all details of your computations, including intermediate values and extrapolated values. **(4p)**

Using Romberg's method, we obtain the extrapolation scheme

$T(1)$	.183 939 721				
		19 887 602			
$T(1/2)$	.243 602 525		.263 490 127		
		5 147 505		46 828	
$T(1/4)$	.259 045 040		.264 192 545	.264 239 373	
		1 298 254		3 034	27
$T(1/8)$	.262 939 802		.264 238 055	.264 241 089	.264 241 117

The last correction is $2.7 \cdot 10^{-8}$ and is clearly smaller than the desired accuracy, so the integral can be approximated by 0.264 241 116.

D. Newton's method

1. We are going to solve the the nonlinear system

$$\begin{aligned} x - y - 3 &= 0 \\ 2x + y - 6 \sin y - 6 &= 0 \end{aligned}$$

using Newton's method, with $(x^{(0)}, y^{(0)}) = (1, -2)$ as the initial guess.

- a) Construct the Jacobian matrix at the starting point. **(1p)**
- b) Introduce a suitable notation and state clearly what the algorithmic steps are for each Newton iteration. **(1p)**
- c) Perform one iteration of Newton's method from the starting point and compute the approximation $(x^{(1)}, y^{(1)})$. **(2p)**

Solution. The Jacobian is

$$J = \begin{bmatrix} 1 & -1 \\ 2 & 1 - 6 \cos y \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 3.5 \end{bmatrix}.$$

The algorithm is: Step 1, compute Jacobian $J(z^{(k)})$; Step2, compute residual $f(z^{(k)})$; Step 3, solve $J\delta z^{(k)} = -f(z^{(k)})$; Step 4, update approximation, $z^{(k+1)} = z^{(k)} + \delta z^{(k)}$. Repeat until convergence.

Here the residual at the starting point is $f(z^{(0)}) = (0, -0.544)^T$, so we get $\delta z^{(0)} = (0.099, 0.099)^T$ and the improved solution $z^{(1)} = (1.099, -1.901)^T$.

E. Differential equations

1. We want to solve the two-point boundary value problem

$$u'' + u' + u = \sin x$$

with boundary values $u(0) = p$ and $u(1) = q$. Introduce a suitable notation and discretize the problem by a second order method. Construct the system of equations that have to be solved, and make sure to include the boundary conditions. **(3p)**

Solution. We use $\Delta x = 1/(N + 1)$ and put N interior point sin $[0, 1]$. Then use symmetric differences to get 2nd order accuracy:

$$\frac{u_{i-1} - 2u_i + u_{i+1}}{\Delta x^2} + \frac{-u_{i-1} + u_{i+1}}{2\Delta x} + u_i = \sin x_i,$$

where $x_i = i\Delta x$. This equation system holds for $i = 1(1)N$, where we put $u_0 = p$ and $u_{N+1} = q$. Thus we have N equations for the N unknowns, $u_1, \dots, u_N$.

E2. Differential equations — IVPs

1. Consider the ordinary differential equation

$$\ddot{y} = y$$

with initial data $y(0) = 1$, $\dot{y}(0) = 2$.

a) Introduce new variables and rewrite the differential equation as a first order system in standard form. Do not forget to state the initial conditions for the new variables. **(2p)**

b) Compute one step using the explicit Euler method with step size $h = 0.5$. What, in particular, is the numerical approximation to the derivative of y at $t = 0.5$, i.e., what is the approximation to $\dot{y}(0.5)$? **(2p)**

Solution. In a) the system is

$$\dot{y} = z \tag{1}$$

$$\dot{z} = y, \tag{2}$$

with initial conditions $y(0) = 1$ and $z(0) = 2$. Here we seek an approximation to $z(0.5)$. So we take one explicit Euler step of size $h = 0.5$, to get

$$z_1 = z_0 + h \cdot y_0$$

from the second equation. Inserting the initial values gives $z(0.5) \approx z_1 = 2 + 0.5 \cdot 1 = 2.5$.

F. Linear systems

1. We are going to solve $Ax = b$ using the Gauss–Seidel method, which uses the matrix splitting $A = L + D + U$ and iterates according to the formula

$$(L + D)x^{(k+1)} = -Ux^{(k)} + b.$$

Here L is *strictly* lower triangular, U is *strictly* upper triangular, and D is diagonal. Take $x^{(0)} = 0$ as the starting value and perform one iteration on the system

$$\begin{bmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}. \tag{3p}$$

Solution. We have $x^{(0)} = 0$, so we find $x^{(1)}$ by solving $(L + D)x^{(1)} = b$,
or

$$\begin{bmatrix} -2 & & & \\ 1 & -2 & & \\ 0 & 1 & -2 & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix},$$

with solution $x^{(1)} = (-0.5, -1.25, -2.125)^T$.

G. Quick questions (True or false; 0.5p for each correct answer).

1. A large condition number implies that the problem is highly sensitive to small changes in the problem data. **T**
2. A quadrature rule is of order $p = 2$ if it integrates all first degree polynomials exactly. **T**
3. Runge's phenomenon implies that if one takes many interpolation points, the interpolating spline will have very large oscillations. **F**
4. An iterative method for $f(x) = 0$ is cubically convergent if the error $\|x_n - x^*\| \approx \epsilon/n^3$. **F**
5. A singular matrix A always has norm equal to 0. **F**
6. If the linear system of equations $x = Ax + b$ is sparse, then it may be preferable to use Jacobi iterations rather than Gauss elimination. **T**
7. The $N \times N$ matrix $\text{tridiag}(1 \ -2 \ 1)$ can be LU -factorized in only $O(N)$ floating point operations. **T**
8. A matrix with $\det A = 0$ cannot be LU factorized. **F**
9. The optimal way of fitting a straight line to given data is independent of what norm is used to measure the residual. **F**
10. When a numerical time-stepping method is used to solve the initial value problem $\dot{y} = \lambda y$ with $\lambda \in \mathbb{C}$, then stability depends only on $\lambda \cdot \Delta t$, where Δt is the time step. **T**

(5p total)

LYCKA TILL!